

CSCI 5314

Dynamic Models in Biology

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (79.0 / 100)	60.0 / 100

60	90	+41
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Surveys computational and mathematical modeling to illuminate biological processes. Students work together to learn to build and analyze models using a variety of numerical tools, tackle meaningful biological problems, and communicate effectively across disciplines. Specific topics: Langevin dynamics of protein folding, agent-based models, finite difference models of organismal growth, stochastic and deterministic cellular automata game of life, models of behavior.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	79.0	100.0	+21.0	MEDIUM
Data Scientist	65.6	100.0	+34.4	HIGH
ML Engineer	61.8	80.2	+18.4	MEDIUM
Machine Learning Engineer	58.5	91.1	+32.6	HIGH
AI Engineer	58.1	92.9	+34.8	HIGH
MLOps Engineer	37.2	78.2	+41.0	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Business Communication

2 of 6 roles

insights

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1** **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2** **MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

GAP KEYWORDS BY ROLE

Applied Scientist

79.0 → 100.0 (+21.0 pts)

biology research diffusion agents data learning systems design

Data Scientist

65.6 → 100.0 (+34.4 pts)

rules data sets insights social approaches

ML Engineer

61.8 → 80.2 (+18.4 pts)

data learning machine systems model impact form building

Machine Learning Engineer

58.5 → 91.1 (+32.6 pts)

data sets learning business design development science model

AI Engineer

58.1 → 92.9 (+34.8 pts)

data learning form software model systems based platform

MLOps Engineer

37.2 → 78.2 (+41.0 pts)

detection data model text systems learning machine image

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.